Draft - expert review pending

Note: This unit is complete and passes the platform's automated checks, but it is not reviewed yet. So, treat its sources and claims with care. You are encouraged to report errors and provide feedback.

Random Variables and Distributions

Unit 9 gave you the probability of a single event, such as "a pupil is absent today". This unit, which follows Chapter 10 of the course guide, takes the next step. Instead of one event, we look at a number that changes by chance, such as *how many* pupils are absent today, and we describe every value that number can take together with its probability. That description is called a probability distribution, and it is the bridge between probability and the statistics you will meet in the rest of the course.

Unit learning outcomes

By the end of this unit you will be able to:

1. Explain what a random variable is, and classify a random variable as discrete or continuous with a reason.
2. Build the probability distribution of a discrete random variable from records or from equally likely outcomes, and check whether a table is a valid probability distribution.
3. Compute the expected value of a discrete random variable and interpret it as a long-run average, not as the most likely value.

Prerequisite knowledge

Unit 9 (basic probability and the probability rules). Unit 1 (discrete and continuous data). Unit 4 (the mean), because the expected value is a

mean.

In this unit

1. Random variables

   

   textbook.com.pk/semester-2/gqur-301/unit-10/topic-01

   - numbers that come from chance, and the difference between counting and measuring.

2. Probability distributions

   

   textbook.com.pk/semester-2/gqur-301/unit-10/topic-02

   - listing every value with its probability, checking the table, and finding the expected value.

How to use this unit

Each topic is one learning cycle. Start with the classroom situation, work through the explanation, and try the activity before you move on. The "Check your understanding" items help you retrieve what you have learned: try them without looking back, then check. Keep a calculator nearby for Topic 10.2. The unit ends with a summative assessment that draws both topics together.

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